

1 Question (15 points)

Given the input alphabet $\{0, 1\}$. Give state diagram of DFA of $L = \{w \mid w \text{ contains an even number of 0s, or contains exactly two 1s}\}$

2 Question (15 points)

Given the input alphabet $\{0, 1\}$. Give state diagram of NFA of $L = \{w \mid w \text{ contains one or two } 0\text{'s in the first four symbols}\}$

3 Question (15 points)

Is $A = \{1^{3^n} \mid n \geq 0\}$ regular? Prove or disprove.

4 Question (15 points)

Let $\Sigma_2 = \left\{ \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \end{bmatrix} \right\}$

Here, Σ_2 contains all columns of 0s and 1s of height two. A string of symbols in Σ_2 gives two rows of 0s and 1s. Consider each row to be a binary number and let

$C = \{w \in \Sigma_2^* \mid \text{the bottom row of } w \text{ is three times the top row}\}$. Show that C is regular. Hint: Work with reverse of w , and remember that if a language is regular then so is its reverse.

For example, $\begin{bmatrix} 0 \\ 1 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \in C$, $\begin{bmatrix} 0 \\ 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \end{bmatrix} \in C$, $\begin{bmatrix} 0 \\ 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \in C$